

**Paper 7**

**AI-Assisted Portfolio Triage for Canada's Major Projects: Expected Value and Developer Break-Even Thresholds**

**Running Title:** AI Portfolio Triage – EV and Break-Even Metrics

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**Keywords:** Major Projects Inventory; Expected Value; Developer Engine; PV-MOIC; Break-Even Multiple; infrastructure policy; project finance

**Word Count:** ~3,400

**Date:** September 2025

**Funding Statement:** This research received no external funding.

**Conflict of Interest:** The author declares no conflict of interest.

**Abstract**

This paper applies a combined Expected Value (EV) and Developer Engine framework to Canada's 2024 Major Projects Inventory (MPI). The EV analysis estimates both the scale and timing of potential investment by distributing project success probabilities across 2025–2027. Results show a heavily front-loaded pipeline, with \$77 billion and 65 expected FIDs in 2025, declining to \$28 billion and 28 FIDs in 2026 and \$23 billion with 23 FIDs in 2027. Investment is concentrated in Alberta (\$65B), Ontario (\$29B), and Quebec (\$16B), and dominated by clean electricity (\$47B), upstream oil and gas (\$19B), and CCUS (\$11B). The Developer Engine models development outflows and FID payoffs to derive risk-adjusted break-even multiples. While clean electricity projects in Alberta and Ontario cluster at feasible levels (~2.0), others require extreme thresholds—coal in BC at 374, critical minerals in BC at 229, and downstream oil and gas in BC at 136. Diversification analysis shows Alberta and BC as the most resilient portfolios, while provinces like Newfoundland and Labrador and Quebec remain concentrated. These findings underscore that while headline investment potential is substantial, project viability depends on aligning economics with targeted policy supports. Instruments such as grants, refundable tax credits, and regionally differentiated incentives for development costs can reduce effective multiples in high-risk categories, whereas uniform national policy risks over-incentivizing projects in low-multiple jurisdictions while leaving higher-multiple provinces and sectors commercially unviable.

## Introduction

Canada's Major Projects Inventory (MPI) records approximately 500 large-scale energy, mining, and forestry proposals that collectively represent more than \$600 billion in announced investment. Yet historical experience shows that one-in-four of these projects proceed to construction. This gap between ambition and realization creates uncertainty for policymakers, financiers, and developers, who must allocate scarce resources across a crowded and risky pipeline. Previous work in this series of papers addressed this research problem through progressively refined probabilistic models. Papers 1 and 2 established the descriptive and historical foundations of the MPI, documenting project churn, survival rates and times, and systemic bottlenecks. Paper 3 introduced two statistical approaches, a Bayesian scorecard and a Cox proportional hazards model, to estimate the probability and timing of a project reaching construction. Paper 4 operationalized these models at the portfolio level, producing blended probabilities and urgency-adjusted priority indices. Paper 5 extended probability analysis by developing a composite power ranking, integrating feasibility and urgency into a single triage metric. Paper 6 demonstrated deployment of these methods in a minimum viable product dashboard, transforming the MPI from a static registry into an interactive platform for probability-based project triage.

This paper extends the application of probabilistic weights and portfolio-level ranking to estimate the expected value (EV) of the 2024 MPI portfolio at the point of final investment decision (FID) over the three-year horizon from 2025 to 2027. In doing so, it addresses two critical gaps in the strategic forecasting of major projects. The first is the need to quantify the magnitude and timing of potential investment flows under uncertainty. The second is to identify

the threshold, or risk-adjusted break-even development cost multiple, required to sustain a steady inflow of freshmen projects (those entering the inventory for the first time) and to limit dropouts (projects that exit before reaching FID).

The contribution of this work is therefore twofold. Methodologically, it integrates probabilistic triage with discounted cash flow analysis in a unified modeling framework, linking project-level risk to financial viability metrics. Practically, it delivers a tool that supports both strategic planning at the portfolio level and economic assessment at the pre-construction stage. By integrating EV analysis with developer break-even thresholds, the model enables stakeholders to map the three-year investment landscape and anticipate when and where developers will commit capital, driven by risk-adjusted returns rather than policy mandates or national interest.

## **Methods**

This study applies a unified yet modular modeling framework to meet two complementary objectives. The first, the EV Engine, assigns probabilities and calculates expected values to estimate the EV of a 2024 MPI project, or a portfolio of projects, at final investment decision (FID) across the 2025–2027 horizon. The second, the Developer Engine, builds on those probabilities to calculate the risk-adjusted break-even multiple of development costs, specifying what developers must recover at FID after accounting for both the time value of money and the risk of project failure.

### *EV Engine*

The EV Engine generates probabilistic estimates of portfolio value at FID. Project-level probabilities of reaching FID are sourced from the output of the Bayesian and Cox risk models

(blended probability by default). For each project, the three-year total probability mass is distributed across 2025, 2026, and 2027 using an urgency-scaled “softmax” function that converts real-valued scores into a probability distribution. The two parameters are years remaining and urgency. This ensures that higher urgency and shorter timelines shift probability weight toward earlier years.

Each project is assigned a value proxy. The default proxy is capital expenditure (project size), though the model can substitute alternatives such as unit count. Expected value for each year is calculated as the product of annual probability and the proxy amount, producing EV\_2025, EV\_2026, and EV\_2027. Cumulative EV is the sum across years, while discounted EV expresses values in 2024 terms using a configurable discount rate. By default, this rate is set to 0 percent, keeping EV as a pure, undiscounted statistical measure of portfolio potential. This choice isolates probability-weighted value from financial assumptions, providing a transparent baseline for strategic planning.

A market-based rate, such as 13 percent (a prime rate of 5 percent plus 8 percent), is a reasonable alternative. Applying such a rate aligns the EV calculation with financial realities and project risk profiles, consistent with the developer-side metrics discussed below. It also opens a path for future research, where EV could be transformed into a fully risk- and time-adjusted measure of portfolio value from an investor’s perspective.

In either case, the EV Engine yields both annual and cumulative measures of expected investment flow, offering a strategic view of the magnitude and timing of potential capital commitments under uncertainty.

*Developer Engine*

The Developer Engine builds on the EV analysis by calculating the risk-adjusted break-even multiple of development costs required to make projects feasible from a developer's perspective. Because the MPI does not record development cost (DevCost), it is assumed as a percentage of total project cost (defaulted to 3 percent but configurable) representing expenditures incurred pre-FID. Because both inflows and outflows are expressed relative to DevCost, the specific percentage chosen (e.g., 3 percent versus 5 percent) does not materially affect the results, ensuring the analysis remains robust to this assumption.

Developer outflows are modeled as an S-curve distributed from the project start year through the pre-FID period, reflecting the gradual but accelerating pattern of development spending.

Developer inflows are modeled as a single payment at FID equal to a multiple of DevCost. For instance, a multiple of one represents simple reimbursement of DevCost, while a multiple of three represents a payment equal to three times DevCost.

The Developer Engine discounts outflows and inflows to a 2024 base year using a 13 percent rate, consistent with a weighted average cost of capital assumption of a 5 percent prime rate plus 8 percent, though this rate is configurable. It then applies the annual FID probabilities generated by the EV Engine to probability-weight the adjusted cash flows year over year, yielding three key performance measures.

The first measure is the multiple of DevCost,  $k$ , when the developer receives only a reimbursement of DevCost at FID. It is calculated by dividing the present value of inflows by the present value of outflows. Under positive discounting assumptions with pre-FID outflows and a later inflow, the resulting multiple is typically less than 1. This signifies that reimbursement alone does not fully compensate the developer for the time value of money or project risk.

The second measure is the break-even multiple,  $k^*$ , defined as the minimum multiple of DevCost required for inflows to equal outflows in present-value terms. Unlike the first measure, where  $k < 1$  indicates a shortfall,  $k^* > 1$  shows how much more than reimbursement must be recovered for developers to break even. By construction, the two measures are reciprocals ( $k^* = 1 \div k$ ), together framing the gap between under-compensation at reimbursement and the hurdle required for feasibility. For example, if  $k = 0.8$ , then  $k^* = 1.25$ , meaning the developer must recover 25 percent more than simple reimbursement to reach break-even, accounting only for the time value of money.

The third, and focal output, builds on the first two by incorporating project risk directly. It is the risk-adjusted break-even multiple,  $k^*_{adj}$ , defined as  $k^* \div \text{blended probability}$ . This measure reflects both the time value of money and the likelihood of project failure. For example, if  $k^* = 1.25$  and the blended probability of success is 25 percent, then  $k^*_{adj} = 5$ . In practice, this means the developer payment at FID must equal five times DevCost to compensate not only for cash flow discounting but also for the risk that three out of four projects fail to reach completion.

### *Integration*

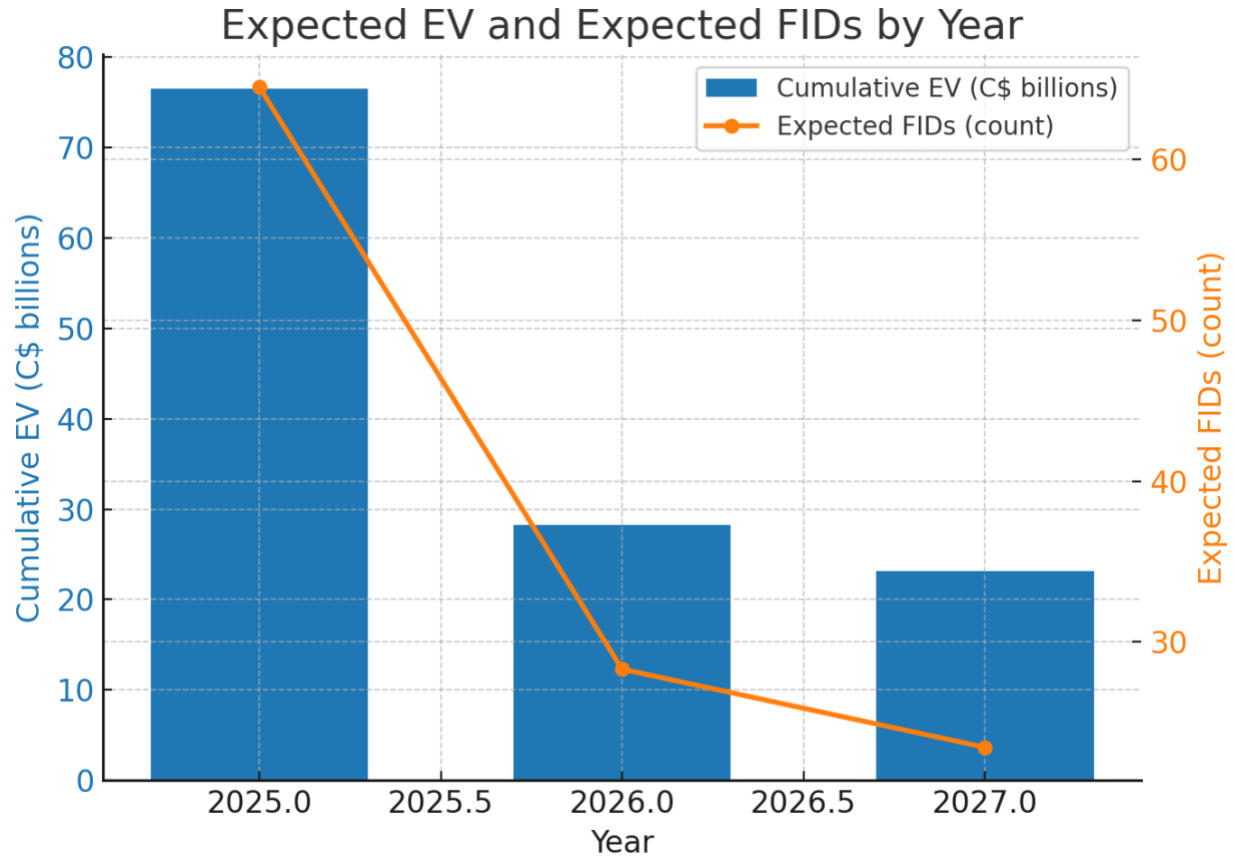
The EV Engine and the Developer Engine are implemented within a single script to ensure consistency and replicability. Because the Developer Engine relies on the annual probability allocations generated by the EV Engine, the two are complementary rather than independent. This integration enables simultaneous reporting of portfolio-level expected values and developer-level payoff thresholds. Conceptually, the EV Engine provides a strategic planning lens for policymakers and financiers, while the Developer Engine offers a viability lens for developers. Together, they deliver a dual perspective on the MPI portfolio. EV highlights the scale and

timing of expected investment flows, while the risk-adjusted break-even development cost multiple explains the minimum condition under which economically rational private actors will commit resources and, ultimately, what is needed to sustain a steady inflow of freshmen projects, limit dropouts, and raise graduation rates into construction.

## **Results and Discussion**

Figure 1 presents cumulative EV in billions of Canadian dollars alongside expected counts of FIDs over the three-year period, 2025–2027. The distribution is heavily front-loaded, with about \$77 billion of cumulative EV and 65 expected FIDs in 2025. This declines to roughly \$28 billion and 28 FIDs in 2026, and \$23 billion with 23 FIDs in 2027. Expected FIDs represent the probability-weighted sum of annual FID probabilities across all projects, yielding an expected rather than literal count per year.

The overall pattern reflects the urgency-weighted allocation in the model, which shifts probability mass toward earlier years when development windows are shorter. In practical terms, this suggests a surge of near-term decision activity that could strain permitting, financing, and infrastructure capacity in 2025. At the same time, there is uncertainty regarding these estimates, so delays in approvals, policy changes, market shifts, or supply-chain constraints could redistribute outcomes into later years.



*Figure 1: Expected EV and Expected FIDs by Year (2025 to 2027).*

Figure 2 turns to geography and sector, showing cumulative EV across provinces and sectoral groups. Alberta dominates with approximately \$65 billion of expected value, followed by Ontario at \$29 billion and Quebec at \$16 billion. At the sectoral group level, clean electricity leads with about \$46 billion, while oil and gas upstream accounts for \$19 billion, and CCUS contributes \$11 billion. Other groups such as low-carbon fuels and waste biomass contribute more modest totals. The results highlight the dual structure of the portfolio. Legacy energy and resource projects still anchor the largest capital flows, but clean electricity is emerging as a central component of expected investment. For planners, the implication is twofold. On one hand, permitting and infrastructure readiness in Alberta, Ontario, and Quebec will

disproportionately shape national outcomes. On the other hand, diversifying support toward smaller jurisdictions and newer technologies could broaden participation and reduce concentration risk.

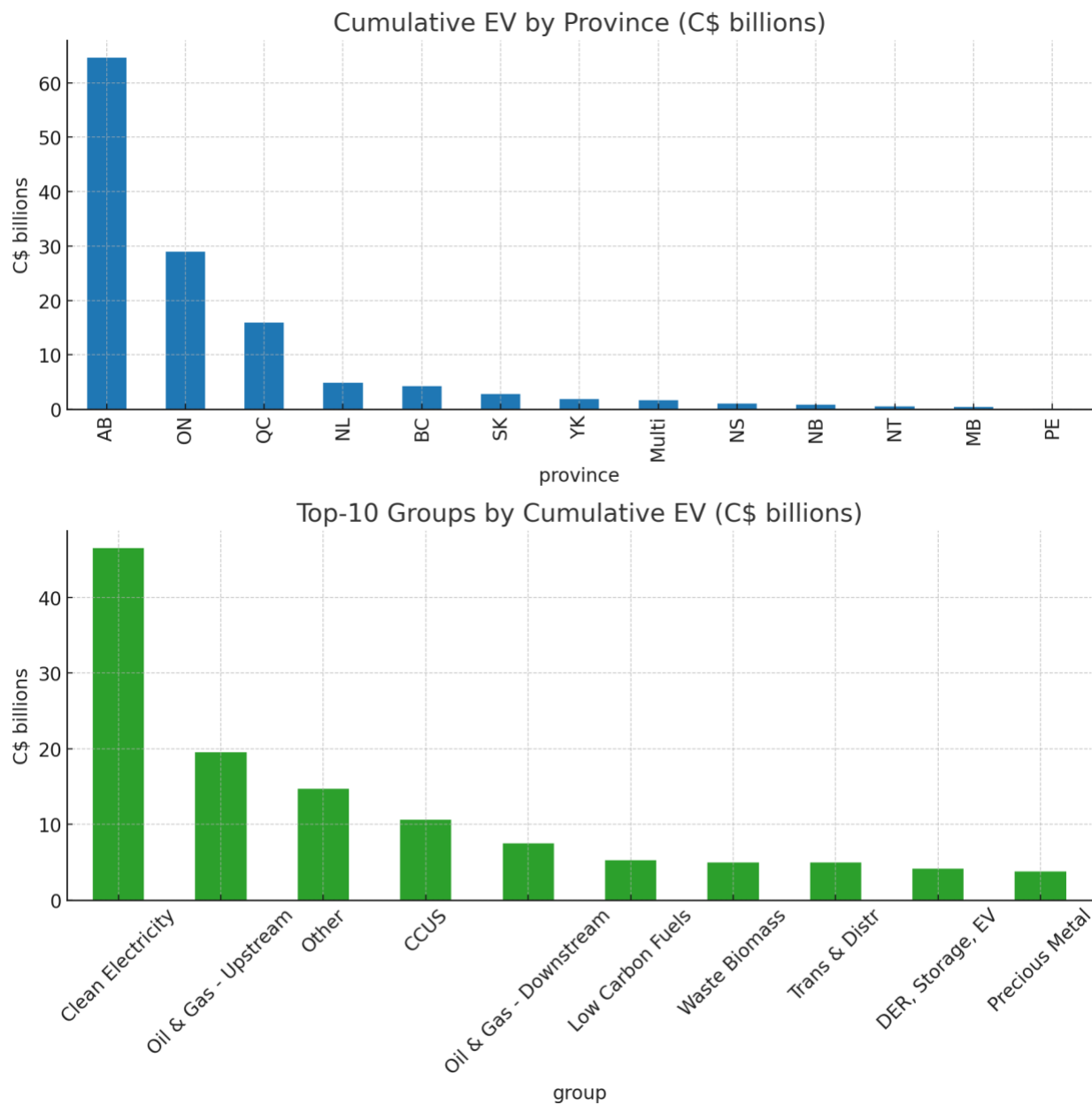
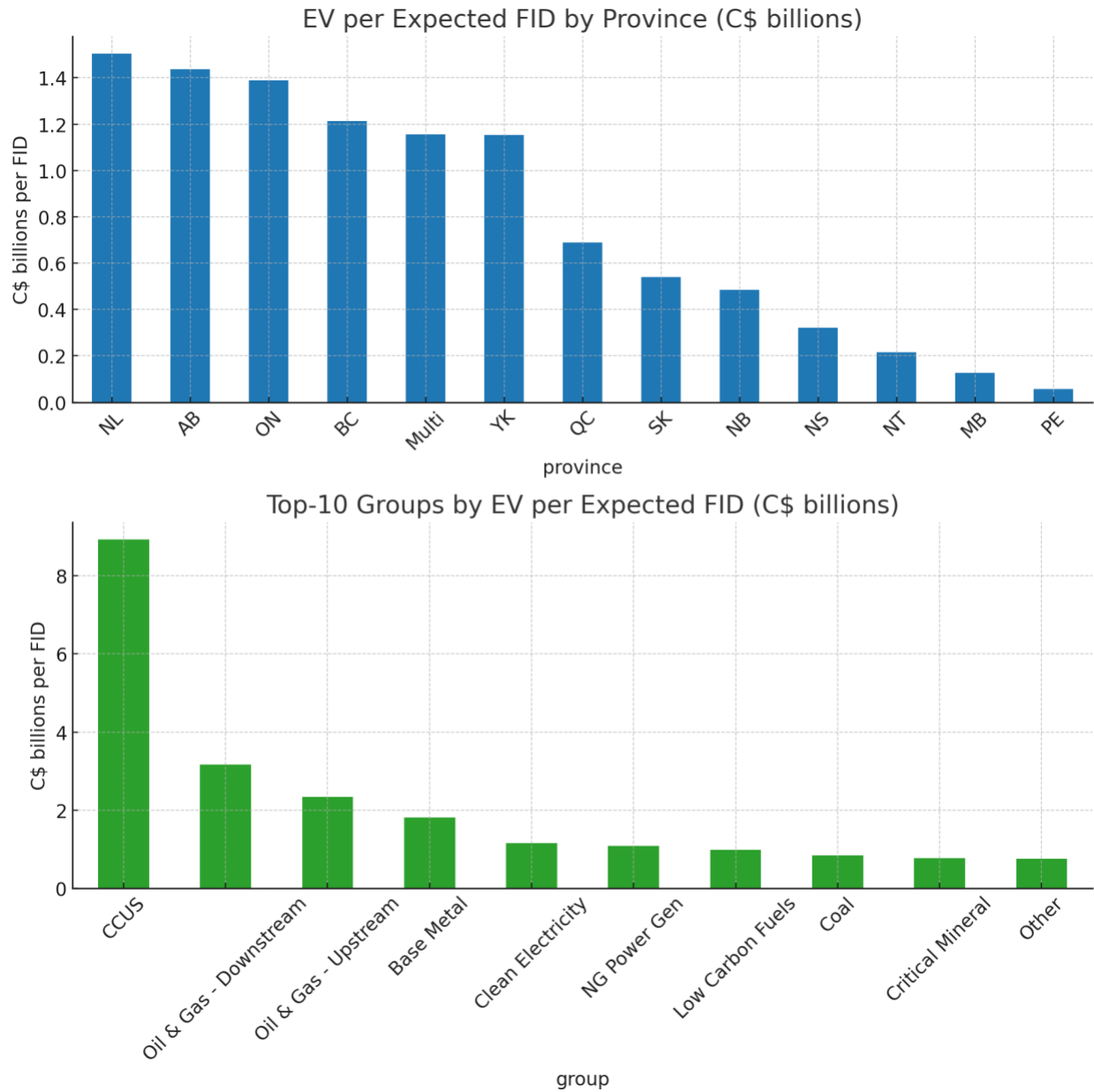


Figure 2: Cumulative EV by Province and Top 10 Group.

Figure 3 scales EV relative to expected FIDs, effectively capturing the average “ticket size” of a decision. By province, Newfoundland and Labrador records the highest average at about \$1.5 billion per FID, reflecting a portfolio concentrated in a few very large, capital-intensive projects. Alberta and Ontario follow closely, each at roughly \$1.4 billion per FID. Smaller provinces such as Prince Edward Island and Manitoba have averages well under \$0.2 billion. By group, CCUS projects stand out, with an average of \$8.9 billion per FID, an order of magnitude higher than most other categories. Oil and gas downstream follows at \$3.1 billion, and oil and gas upstream at \$2.4 billion. Clean electricity, while large in total, averages about \$1.1 billion per FID, reflecting a higher number of projects. This framing reveals that not all decisions carry equal weight. In some cases, a single outcome could reshape provincial or sectoral trajectories, while in others the stakes are smaller and more distributed.



*Figure 3: Average EV per Expected FID by Province and Top 10 Group.*

Figure 4 presents the same scaling using the median rather than the mean, highlighting the “typical” project size reaching FID rather than the portfolio average. By province, the outlier effect of multi-jurisdictional projects is muted, while provinces such as Saskatchewan, Ontario, and Alberta show median values in the \$0.4–0.6 billion range. Smaller provinces cluster even lower, often under \$0.2 billion per FID. By group, CCUS remains the leader, with a median

close to \$9 billion per FID, but the gap with other categories narrows. Oil and gas downstream and upstream projects fall in the \$1–3 billion range, while clean electricity and natural gas power generation show typical project sizes below \$1 billion. Using the median reveals that while a few mega-projects can dominate averages, most decisions involve smaller, more representative investments, providing a clearer sense of what a “typical” FID looks like across provinces and sectors.

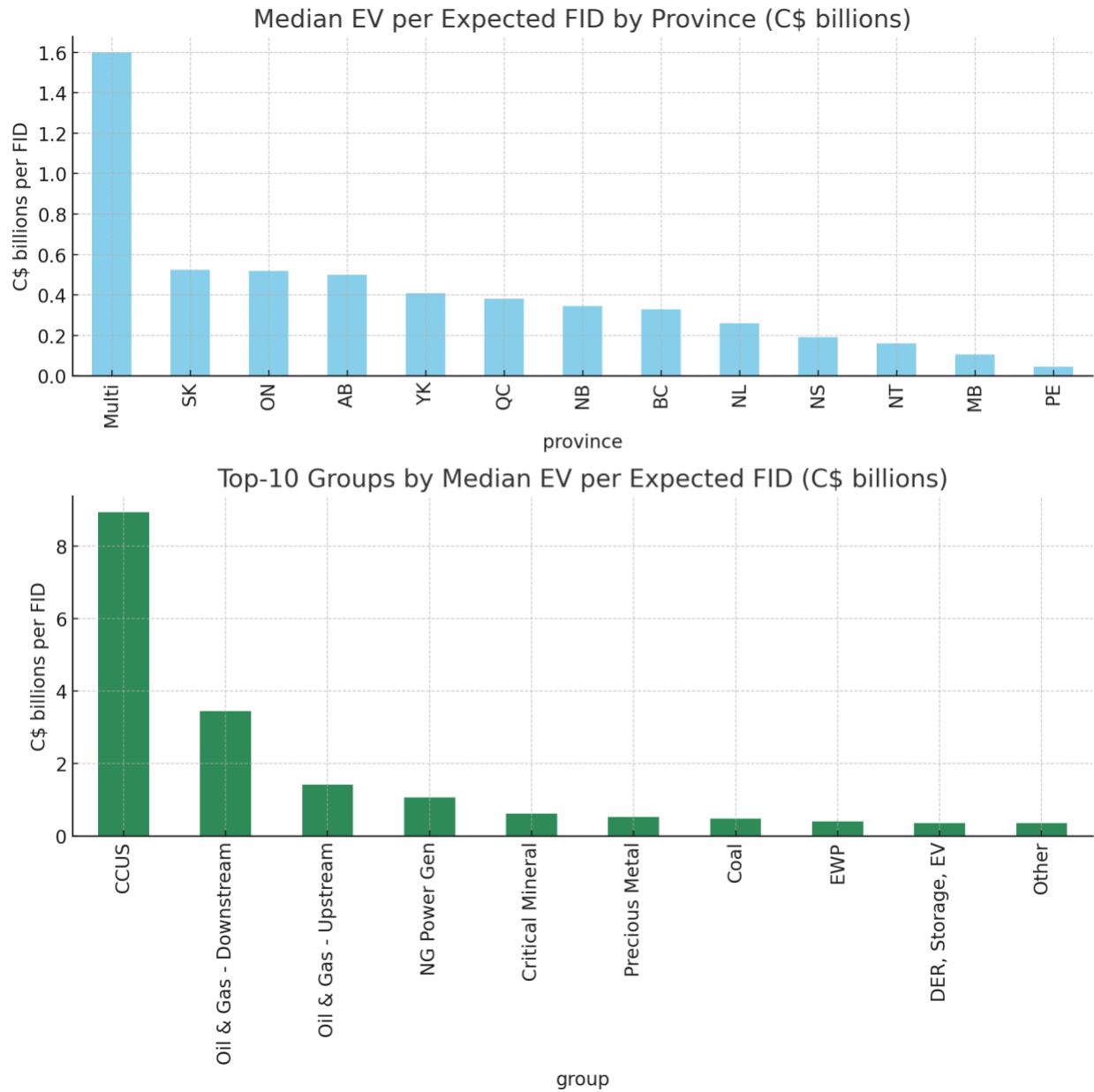


Figure 4: Median EV per Expected FID by Province and Top 10 Group

Figure 5 provides a heatmap of cumulative EV by group and province. A small number of combinations dominate: oil and gas upstream in Alberta (\$19 billion); clean electricity in Ontario (\$17 billion), Alberta (\$16 billion), and Quebec (\$12 billion); and CCUS in Alberta (\$11 billion). Other notable contributions include “Other” projects in Alberta (\$7 billion) and distributed energy and storage in Ontario (\$3 billion). Most other cells register less than \$2 billion. The

concentration of expected value into these few intersections underscores the uneven structure of the pipeline. It also highlights potential bottlenecks, as the success or failure of a handful of sectors in a few provinces could drive portfolio-level outcomes.

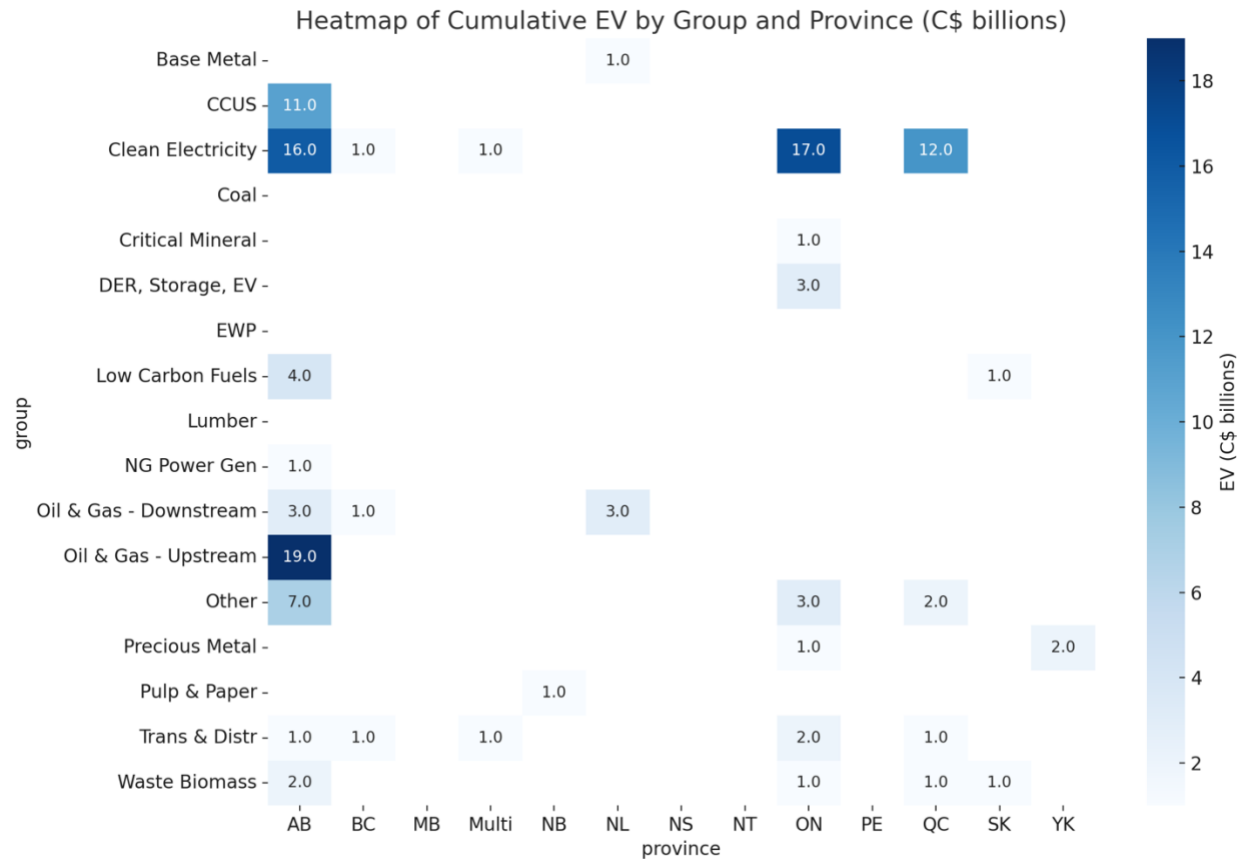
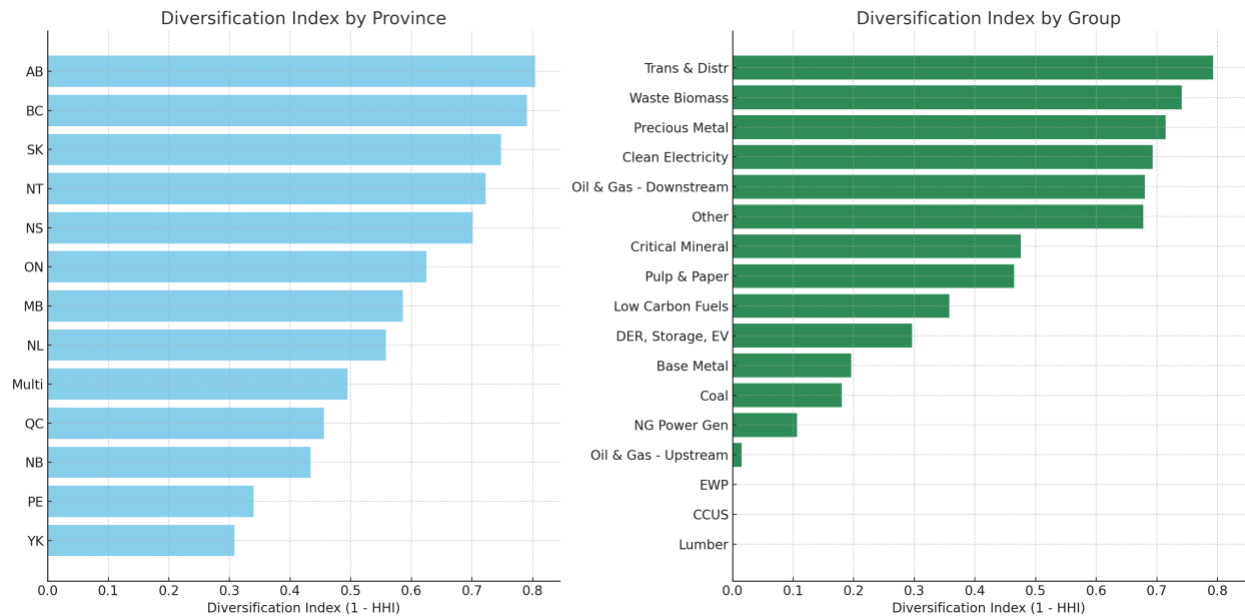


Figure 5: Heatmap of Cumulative EV by Group and Province

Figure 6 presents an EV diversification index by province and by group. The provincial index measures how widely a province's cumulative EV is distributed across project groups, with higher values indicating greater sectoral diversification and resilience. The group index measures how widely a group's cumulative EV is distributed across provinces, with higher values indicating greater geographic diversification and resilience. The index is based on the Herfindahl–Hirschman Index (HHI), calculated as the sum of squared EV shares, across groups

within each province and across provinces within each group. The Diversification Index is defined as  $1-HHI$ , where 1 indicates perfect diversification and 0 indicates full concentration in a single category.



*Figure 6: Diversification Index by Province and Group*

**Most Diversified Provinces.** Alberta (0.80) shows the broadest spread, with EV distributed across oil & gas upstream, downstream, CCUS, clean electricity, and other categories. British Columbia (0.79) has a balanced portfolio, combining clean electricity, CCUS, upstream oil & gas, and smaller categories. Saskatchewan (0.75) is diversified across clean electricity, transmission, and resource projects.

**Most Diversified Groups.** Transmission & Distribution (0.79) is the most geographically balanced, followed by Waste Biomass (0.74) and Precious Metals (0.71). Clean Electricity (0.69) is also broadly diversified, with projects spread across Alberta, Ontario, Quebec, BC, and Atlantic provinces.

**Moderately Diversified Provinces.** Northwest Territories (0.72) achieves balance across smaller-scale projects. Nova Scotia (0.70) demonstrates a reasonably even spread but at lower overall EV. Ontario (0.62) shows significant EV concentration in clean electricity, though it still spans several groups. Manitoba (0.59) is moderately diversified but with fewer large categories.

**Moderately Diversified Groups.** Oil & Gas – Downstream (0.68) shows moderate diversification, reflecting activity in multiple provinces. “Other” projects (0.68) and Critical Minerals (0.48) are spread across several regions but remain concentrated in a few provinces. Pulp & Paper (0.46) shows a similar profile, anchored in Ontario and Quebec but with smaller activity elsewhere.

**Least Diversified Provinces.** Newfoundland & Labrador (0.56) is dominated by a few large, capital-intensive projects, leaving it exposed to single-project outcomes. Multi-jurisdictional projects (0.49) appear diversified by geography but are concentrated in a narrow set of categories. Quebec (0.46) is heavily tilted toward clean electricity. New Brunswick (0.43) is similarly concentrated, with limited sectoral variety. Prince Edward Island (0.34) and Yukon (0.31) are the least diversified, dependent on single groups with little breadth.

**Least Diversified Groups.** Low Carbon Fuels (0.36) are concentrated mainly in Alberta, while DER, Storage, and Electric Vehicles (0.30) remain niche and provincial. Base Metals (0.20) and Coal (0.18) are concentrated in limited geographies. Natural Gas Power Generation (0.11) and Oil & Gas – Upstream (0.02) are highly concentrated, with the latter almost entirely in Alberta. CCUS (0.00), Engineered Wood Products (0.00), and Lumber (0.00) are the least diversified, dominated by single-province portfolios.

Figure 7 shifts the focus from strategic planning to developer economics, showing the risk-adjusted break-even development cost multiple ( $k^*_{adj}$ ) by group and province. Values vary widely. Coal projects in British Columbia require a median multiple of 374.3, and critical minerals in the same province 228.9. Oil and gas downstream also shows elevated requirements in British Columbia, with a median of 136.1. By contrast, many mainstream combinations, such as clean electricity in Ontario and Alberta, cluster between 1.8 and 2.0. These differences reflect the interplay of risk, timing, and project mix.

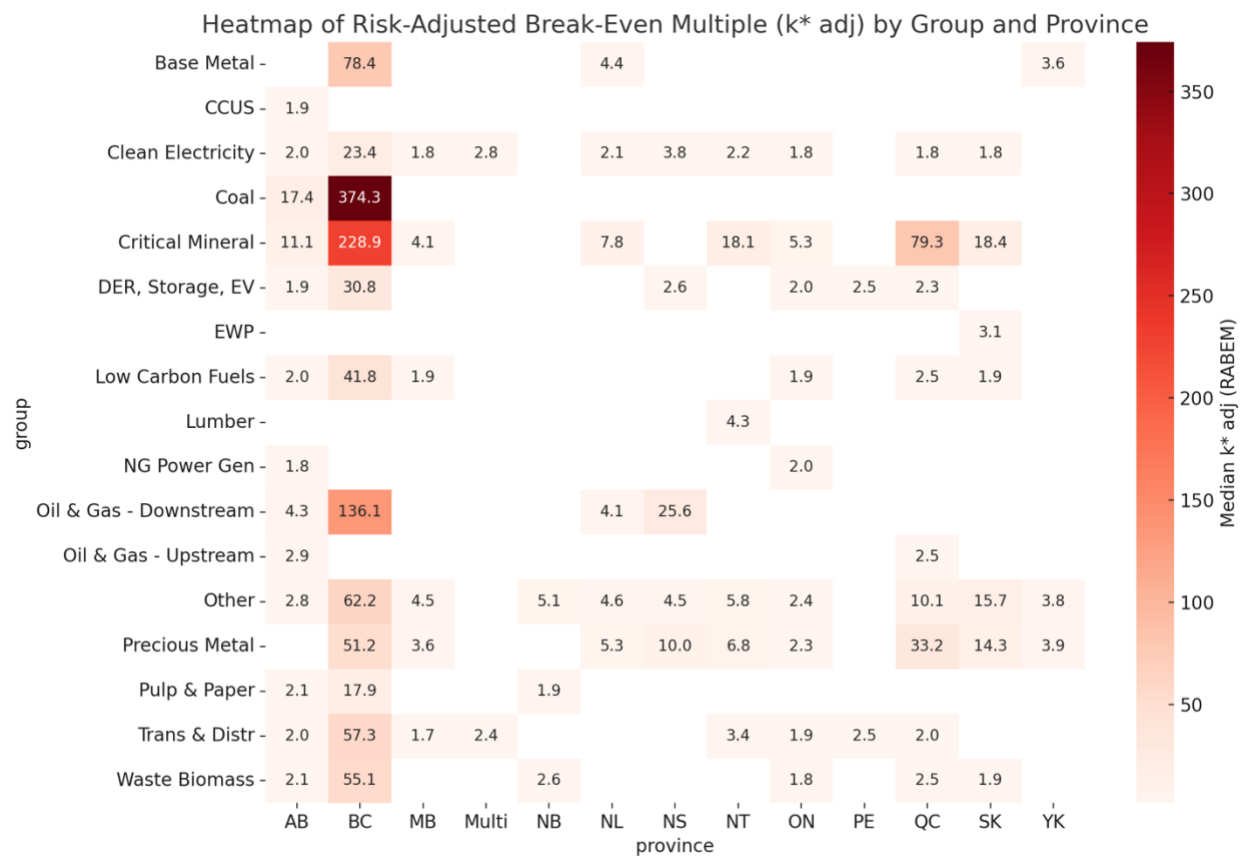


Figure 7: Heat Map of Risk-Adjusted Break-Even Multiple by Group and Province.

For developers, provinces or sectors with high multiples signal environments where projects are unlikely to proceed without extraordinary returns or strong policy support. In these cases, grants

and other non-repayable public supports become the primary policy lever for lowering effective multiples to investable levels. For policymakers, the implication is that incentive frameworks may need to be regionally and sectorally differentiated. A uniform national policy risks over-incentivizing projects with inherently low multiples while leaving higher-multiple jurisdictions or sectors commercially unviable.

For sake of clarity, it is important to note that the multiple does not imply a project sponsor could or would divest the project at FID. Rather, it represents part of the decision calculus for committing at-risk development expenditures, whether with the expectation of monetization at FID or of advancing the project internally to the capital investment stage. In practice, the multiple is seldom realized as a discrete, one-time payout. Instead, it is generally capitalized into overall project costs, with higher multiples translating into elevated internal rate of return thresholds and required hurdle rates, all else being equal.

Further research into this area could deepen the sectoral and regional analysis of payoff thresholds, refine probability models with updated policy and market data, and explore the dynamic interaction between public incentives and private capital allocation. Extending the framework to international comparisons could also shed light on Canada's relative competitiveness in attracting investment.

## **Conclusion**

This analysis applied a combined EV and Developer Engine framework to Canada's 2024 major project pipeline. The EV component estimated the scale and timing of potential investment by distributing three-year project success probabilities across 2025–2027 using urgency-scaled softmax weights. This approach captures how more urgent projects with shorter timelines are

more likely to advance earlier, while longer-horizon projects spread probability further out. EV was then expressed in dollar terms by multiplying these annual probabilities by project costs, yielding both annual and cumulative expected values.

The Developer Engine quantified required payoffs by modeling development costs, S-curve outflows, and FID inflows adjusted to a 2024 base year. From these probability-weighted streams, a risk-adjusted break-even development cost multiple was derived to reflect the effective payoff once time value of money and project risk are incorporated. Importantly, this multiple does not imply a sponsor could sell a project at FID, but rather forms part of the calculus in deciding whether to commit at-risk development dollars, either in anticipation of a sale or to advance an internal capital project. In practice, higher multiples tend to be absorbed into overall project returns, raising required hurdle rates and internal rate of return thresholds.

The results highlight sharp variation across provinces and sectors. Strategically, EV is heavily concentrated in Alberta (\$65 billion), Ontario (\$29 billion), and Quebec (\$16 billion), with clean electricity (\$47 billion), upstream oil and gas (\$19 billion), and CCUS (\$11 billion) dominating by sector group. Yet diversification analysis shows these headline figures mask uneven resilience. Alberta and BC emerge as the most diversified provinces, with EV spread across hydrocarbons, CCUS, and renewables, while provinces such as Newfoundland and Labrador and Quebec are dominated by a few large categories. Similarly, groups such as transmission and distribution and clean electricity are geographically widespread, whereas upstream oil and gas and CCUS is highly concentrated in a single province. This unevenness suggests that shocks in a handful of categories could reverberate across the national portfolio.

Economically, payoff thresholds diverge just as sharply. While mainstream combinations such as clean electricity in Alberta and Ontario cluster at realistic multiples (around 2.0), others show extreme requirements—coal in BC at 374, critical minerals in BC at 229, and oil and gas downstream in BC at 136—that are unlikely to be achieved without extraordinary returns or targeted support. This juxtaposition of expected investment value, diversification patterns, and viability metrics underscores that Canada’s headline potential will not automatically translate into projects under construction.

Success will depend on aligning project economics with targeted policy supports. Instruments such as grants, refundable tax credits, and regionally differentiated incentives for development costs can reduce effective multiples in high-risk categories, whereas a uniform national policy risks over-incentivizing projects in low-multiple jurisdictions while leaving higher-multiple provinces and sectors commercially unviable.